basic feasible solution. Here m is the number of rows and n is the number of columns in a transportation problem.

10.1.2 Methods of finding initial Basic Feasible Solutions

There are several methods available to obtain an initial basic feasible solution of a transportation problem. We discuss here only the following three. For finding the initial basic feasible solution total supply must be equal to total demand.

(i.e) $\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$

Method:1North-West Corner Rule (NWC)

It is a simple method to obtain an initial basic feasible solution. Various steps involved in this method are summarized below.

Step 1: Choose the cell in the north-west corner of the transportation Table10.1 and allocate as much as possible in this cell so that either the capacity of first row (supply)is exhausted or the destination requirement of the first column(demand) is exhausted.(i.e) x_{11} = min(a_1,b_1)

Step 2: If the demand is exhausted ($b_1 < a_1$), move one cell right horizontally tothe second column and allocate as much as possible.(i.e) x_{12}= min ($a_1 - x_{11}$, b_2)

If the supply is exhausted ($b_1 > a_1$), move one cell down vertically to the second row and allocateas much as possible.(i.e)x_{21} = min (a_2,$b_1 - x_{11}$)

If both supply and demand are exhausted move one cell diagonally and allocate as much as possible.

Step 3: Continue the above procedure until all the allocations are made

Example 10.1

Obtain the initial solution for the following problem.

		Destination			
		A	B	C	Supply
Sources	1	2	7	4	5
	2	3	3	1	8
	3	5	4	7	7
	4	1	6	2	14
Demand		7	9	18	

Solution:

Here total supply = 5+8+7+14=34,
Total demand = 7+9+18=34

(i.e) Total supply =Total demand. Therefore the given problem is balanced transportation problem.

∴ we can find an initial basic feasible solution to the given problem.

From the above table we can choose the cell in the North West Corner. Here the cell is (1,*A*)

Allocate as much as possible in this cell so that either the capacity of first row is exhausted or the destination requirement of the first column is exhausted.

i.e. x_{11} = min (5,7) = 5

	A	B	C	Supply (a_i)
1	(5) 2	7	4	5/0
2	3	3	1	8
3	5	4	7	7
4	1	6	2	14
Demand (b_j)	7/2	9	18	

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